Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Machine Learning Engineer with 3 years of industry experience building production **retrieval**, **GenAI** and data systems in a regulated financial services setting: **embedding-based semantic search** over **pgvector**, **Python** services and batch workloads on **AWS**, and self-hosted **Llama-3** on a **GPU** instance, with explicit quality, cost and reliability trade-offs. Co-built a multi-signal **scoring and ranking** algorithm for an MSc in Data Science dissertation with AstraZeneca (University of Bristol), refined with scientist feedback. Acted as de facto team lead for up to 4 interns and was twice recognised with Best Team Project (2024, 2025). Seeking to improve search quality through models, data, evaluation and infrastructure at Perplexity.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Retrieval, Ranking & GenAI Systems

- Designed and built production **retrieval** and **GenAI** applications end to end (**ReactJS** frontend, **Python FastAPI** microservices, **RAG** with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & **Llama-3**) serving Compliance, Client Service and Investment, with **Azure OpenAI embeddings** and a separate **pgvector** database per department for **semantic search** over internal documents.
- Built **tool-calling** agents that call **Python** functions to run **SQL**-backed fund comparisons and generate PDF reports on request; versioned every prompt change in **Git** and had the **Compliance** department review each release.
- Made quality-versus-cost trade-offs explicit: **self-hosted Llama-3** on a **GPU EC2** instance for the largest department and kept the stronger **Azure OpenAI** outputs for decision makers.

Serving, MLOps & Production Reliability

- Designed the infrastructure for internal analytics platforms and GenAI applications on **AWS** (**EC2** including a **GPU** instance, **S3**, **ECR** and **Lambda**); packaged the **FastAPI** microservices (**uvicorn**) as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Operated both real-time and batch workloads: user-facing services on **FastAPI** microservices, and a daily rating-email pipeline run on **AWS SageMaker**, scheduled by an **EventBridge** rule that triggers a **Lambda** function at 9 AM.
- Implemented **RBAC** with a **Python** authentication function against the company **LDAP** server; restricted access to company private IPs using **AWS Security Groups**; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

Data Pipelines & Stakeholder Delivery

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data; developed **Optimized SQL** queries, **Functions** and **Procedures**.

- Built a **Python** extraction pipeline that parses over **300** daily emails from credit rating agencies into structured data, **fuzzy-matches** company names against the ones analysts track and sends a daily summary, **saving 60 analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images.
- Analyzed customer data using **Python Pandas** and **Market Basket Analysis** (**Apriori** Algorithm); migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**.
- Worked directly with stakeholders in **Client Service, Compliance and Investment** to turn requirements into working tools; acted as **de facto team lead for up to 4 interns** in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. Data preprocessing pipeline, scoring and ranking algorithm and RAG-based AI assistant for a tool, built as FastAPI microservices with a React frontend, that ranks cancer cell lines by target inclusion and exclusion genes using per-gene scores across six evidence layers, calibrated across lineages; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes and a vision model scoring uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Architecture study comparing CNN/ViT/ResNet encoders and symbolic, SBERT embedding-retrieval and LSTM/attention decoders for generating image descriptions, scored with SBERT cosine similarity, BLEU and ROUGE-L and benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator. (A self-driving car trained using a **Convolution Neural Network (CNN)** model.)

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Azure OpenAI, LlamaIndex, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite